FOR COMPLETE BEGINNERS

A list that can grow.

Complete Modern Java, episode 19: ArrayList, in Java and TypeScript

> Add, read and remove items from a list that changes size as you go.

An array keeps the length it was made with: three boxes, no room for a fourth.

An ArrayList grows when you add and shrinks when you remove.

In this episode

- A full array
- ArrayList<String>
- add, get, size
- remove and contains
- ArrayList<Integer>, not int
- get(5) out of bounds
- TypeScript arrays

Coming next

- A list of products (episode 20)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each run printed

Full.java: basket[3] on a String[3]	`ArrayIndexOutOfBoundsException: Index 3 out of bounds for length 3`
Cart.java: print, size(), get(0)	`[Bread, Milk, Eggs], 3, Bread`
Cart.java: remove("Milk"), contains("Milk")	`[Bread, Eggs], false`
ArrayList<int>	`error: unexpected type, required: reference, found: int`
ArrayList<Integer>: add 150, 99	`[150, 99]`
cart.ts: basket[5]	`undefined (Java get(5) throws)`

Every row was run, not predicted.

02 ArrayList: add, get, size, remove

String between the angle brackets says what the list holds; the empty <> lets Java work it out. No import in a compact source file.

Remove shifts the items after it to the left, so the indexes change.

Cart.java

```
void main() {
    ArrayList<String> basket = new ArrayList<>();
    basket.add("Bread");
    basket.add("Milk");
    basket.add("Eggs");
    IO.println(basket);             // [Bread, Milk, Eggs]
    IO.println(basket.size());      // 3
    IO.println(basket.get(0));      // Bread
    basket.remove("Milk");
    IO.println(basket);             // [Bread, Eggs]
    IO.println(basket.contains("Milk"));   // false
}
```

> Printed exactly these values; basket.get(5) then threw Index 5 out of bounds for length 2.
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

03 Numbers need Integer

A list holds objects. int is a plain number, so ArrayList<int> does not compile; Integer is the object version of int.

Numbers.java

```
void main() {
    ArrayList<Integer> prices = new ArrayList<>();
    prices.add(150);
    prices.add(99);
    IO.println(prices);   // [150, 99]
}
```

> With int: error: unexpected type. With Integer: [150, 99]. remove(1) on [10, 20, 30] left [10, 30].
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

04 TypeScript arrays already grow

push adds, length counts. To remove by value, find the index and check it is not -1 first: splice(-1, 1) removes the last item.

cart.ts

```
const basket: string[] = [];
basket.push("Bread");
basket.push("Milk");
basket.push("Eggs");
const index = basket.indexOf("Milk");
if (index !== -1) {
    basket.splice(index, 1);
}
console.log(basket);       // [ 'Bread', 'Eggs' ]
console.log(basket[5]);    // undefined
```

> tsc was quiet; node printed the lists, 3, -1 and undefined.
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

Use a list

1. It must grow? ArrayList, not an array.
2. size() counts; get(i) reads.
3. Numbers: ArrayList<Integer>.
4. TypeScript: check indexOf is not -1 before splice.
5. Challenge: add Bread, Milk, Eggs; remove Milk; add Rice. size() and get(1)?

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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